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Put and Call Options

Underlying Asset (Assume Stock)

Expiration Date or Maturity T

Strike Price K

Calls: Give holder right, but not the obligation to buy one share of stock at price K .

Puts: Right, but not obligation to sell one share at K .

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European: Can be exercised only at time T . (Exercise means buy for a call, sell for a put.)

American: Can be exercised any time between 0 and T . (Can be exercised at most once.)

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If all parameters and underlying are same, an American Option is worth at least as much as its European Counterpart.

Most Exchange Traded options in US are American.

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Bermudan: Can be exercised on
Some specified set of dates
between 0 and T.

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Options can be used to speculate and also to hedge risk.

Protective Put

Suppose I own 300 shares of AAPL now. (\$148.95/share)
Market Value of stock is
\$44,685

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Suppose I need \$42,000 on 9/24.

Buy Puts with $K = 140$ and $T = 9/24$ (300 Puts)

300 Shares + 300 Puts
worth $\geq \$42,000$ until 9/24

	Bid	Ask
Puts $K = 140, T = 9/24$	1.51	1.55

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For a total of \$465 ($300 \times \1.55)

I can insure my portfolio to be worth at least \$42,000

Apple below 140 → exercise puts
sell shares for 42,000

Apple above 140 → don't exercise
puts

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Other Strikes, Same T

K = 135 (40,500) Bid Ask

.89 .92

K = 149 Bid Ask

4.25 4.35

K = 160 Bid Ask

11.90 12.05

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New Example Speculate

Suppose I believe AAPL $\uparrow$ (soar)

Have \$44,685 in Cash

Strategy 1: Buy 300 Shares AAPL
at \$148.95 / share

Strategy 2: Buy Calls, $K=149$
 $T=9/3$

$C_0 = 2.89$ (ask)

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Buy \$44,685 of calls

15,462 Calls

Scenario 1: AAPL at \$149 on 9/3

Strat #1: 300 Shares 44,700

Strat #2: 300 calls worthless

Scenario 2: AAPL at \$175 on 9/3

Strat #1: 300 Shares 52,500

Strat #2: Each call worth \$26

$$15,462 \times \$26 = \$402,012$$

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Focus on European Options
until further notice

V_t = Price of V at t

Payoff Call

$$C_T = \max(S_T - K, 0)$$

Payoffs $P_{0,t}$

$$P_T = \max(K - S_T, 0)$$

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Call: Valuable when $S_T > K$

Put: Valuable when $S_T < K$

Positive Part Function

$$x^+ = \max(x, 0) = \begin{cases} x, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$$x^+ - (-x)^+ = \begin{cases} x - 0 = x & x \geq 0 \\ 0 - (-x) = x & x < 0 \end{cases}$$

$$X^+ - (-X)^+ = X$$

Put-Call Parity

No Dividends between $0, T$

European Options Same Stock

Same Maturity T

Same Strike K

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Assume borrow or invest at
const. interest r (cont. compounding)

Borrow \$1 at t , repay at $T > t$
Owe at T $e^{r(T-t)}$

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Buy one call, sell one put

Capital at T is

$$C_T - P_T = (S_T - K)^+ - (K - S_T)^+ \\ = S_T - K$$

Set $X = S_T - K$, look at

$$X^+ - (-X)^+ = X$$

Another way to get $S_T - K$
at T

At t : Buy one share, borrow
 $Ke^{-r(T-t)}$

T Own Stock, owe K

Portfolios must have
same value at t

$$C_t - P_t = S_t - Ke^{-r(T-t)}$$

Put-Call Parity

For $t=0$

$$C_0 - P_0 = S_0 - Ke^{-rT}$$

$$C_0 = P_0 \iff S_0 = Ke^{-rT}$$

$$K = S_0 e^{rT}$$

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Which is more expensive
at $t=0$, put or call

$$\text{If } K = S_0 e^{rT} \quad P_0 = C_0$$

$$\text{If } K < S_0 e^{rT} \quad C_0 > P_0$$

$$\text{If } K > S_0 e^{rT} \quad P_0 > C_0$$

Not Valid for
American Options

Put-Call Parity for American Options

Get Inequalities instead
of equations

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Black-Scholes-Merton

Brownian Motion $W_t, t \geq 0$

For each t , W_t is a random
Variable

Each W_t is normal, $W_0 = 0$

For each $t > s$ $W_t - W_s$ normal

Mean 0

Variance $t-s$

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Continuous in t

$$0 \leq s_1 < t_1 \leq s_2 < t_2$$

$W_{t_1} - W_{s_1}$, $W_{t_2} - W_{s_2}$ are
independent

Simulate $t_i = \frac{1}{n}$ $W_{t_0} = 0$

$$\Delta t = \frac{1}{n} \quad t_{i+1} - t_i = \Delta t$$

$$W_{t_{i+1}} - W_{t_i} = \sqrt{\Delta t} \left[\text{Draw from } \mathcal{N}(0, 1) \right]$$

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Stock Price Evolves as

$$dS_t = S_t(\alpha dt + \sigma dW_t)$$

Small Δt

$$S_{t+\Delta t} - S_t = S_t(\alpha \Delta t + \sigma(W_{t+\Delta t} - W_t))$$

$\sigma(W_{t+\Delta t} - W_t)$ Normal

Mean 0

Variance $\sigma^2 \Delta t$

σ - Black-Scholes Volatility Param

$$\mu = \ln S_0 + (r - \frac{1}{2}\sigma^2)T, \quad \nu = \sigma\sqrt{T}$$

$$C = \frac{\ln k - \mu}{\nu}, \quad g(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

$$\overline{\Phi}(x) = \int_{-\infty}^x g(t) dt$$

Yesterday's Math Prep

$$P_0 = Ke^{-rT} \overline{\Phi}(c) - e^{-rT} e^{\mu + \frac{1}{2}\nu^2} \overline{\Phi}(c - \nu)$$

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$$\mu + \frac{1}{2}\sigma^2 = \ln S_0 + rT$$

$$e^{\mu + \frac{1}{2}\sigma^2} = S_0 e^{rT}$$

$$P_0 = K e^{-rT} \Phi(c) - S_0 \Phi(c - \sigma)$$

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Customary to Write

$$d_1 = \frac{1}{\sigma\sqrt{T}} \left(\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right)T \right)$$

$$d_2 = \frac{1}{\sigma\sqrt{T}} \left(\ln\left(\frac{S_0}{K}\right) + \left(r - \frac{1}{2}\sigma^2\right)T \right)$$

$$= \frac{1}{\sigma} (\mu - \ln K) = -C$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

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Exercise: Use Put-Call parity
and P_0 to show

$$C_0 = \Phi(d_1)S_0 - \Phi(d_2)Ke^{-rT}$$

Note: $P_0 = Ke^{-rT}\Phi(-d_2)$
 $- S_0\Phi(-d_1)$

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Implied Volatility

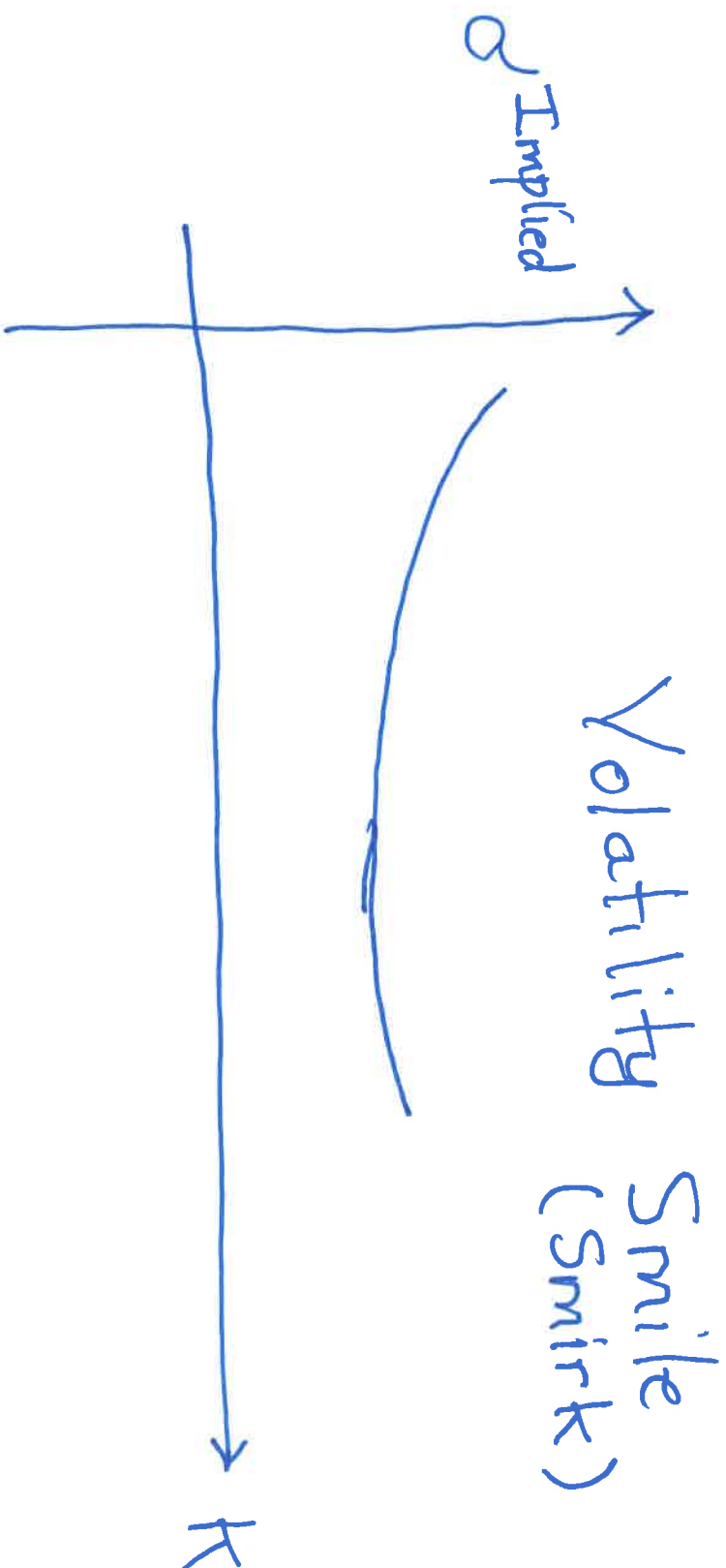
Take Market Price of Option
(Use a Call)

Find σ so that Black-Scholes
Price = Market

Such a σ is called
implied volatility of
option

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If Black-Scholes worked perfectly get same σ_{Implied} for all K, T (for a given stock)



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Early Exercise and American Calls

No Divs $r > 0$ (risk-free rate)
cont. comp.

C_t^E - Eur Call Price

C_t^A - Am. Call Price

Same Strike K

Same Maturity T

For free, we know

$$\textcircled{1} \quad C_t^A \geq C_t^E$$

Let V^E be a European
Security that pays $V_T^E = S_T - K$
at T .

$$C_T^E = (S_T - K)^+ \geq S_T - K = V_T^E$$

$$C_T^E \geq V_T^E \Rightarrow C_t^E \geq V_t^E$$

$$0 \leq t \leq T$$

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Replicate V^E (Portfolio with value at $T = V_T^E$)

At t , buy one stock
borrow $Ke^{-r(T-t)}$

$$V_t^E = S_t - Ke^{-r(T-t)}$$

Fails if
Dividends

$$C_t^E \geq V_t^E$$

$$(2) \quad C_t^E \geq S_t - Ke^{-r(T-t)}$$

If exer Am Call at $t < T$

get $S_t - K$ (Wouldn't
exer if $S_t < K$)

(3)

$$S_t - Ke^{-r(T-t)} > S_t - K$$

Provided $r > 0$
 $t < T$

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$$C_t^A \geq C_t^E \geq S_t - Ke^{-r(T-t)} > S_t - K$$

∴ Price of Amer Call

> what we would get
by exercise

Suppose we have a stock,
current price is 50.

I believe Stock $\nearrow$ in one month

Could buy calls $K=50$, $T=1$ month

Suppose I believe stock price
might be 60 or 65 in one
month, but won't hit 70

Call Spread

Sell Calls $K = 70$

Generates income

If $50 \leq \text{Stock Price} \leq 70$ 1 month

Make Same as without $K = 70$
Calls, but receive money
initially.

Bull Call Spread

Think Stock will go up

Simplistic Strategy Call K, T

Pay $C_{0,K,T}$. Need

$S_T - K > C_{0,K,T}$ to profit

Two Calls, Same T

Strikes $K_1 < K_2$

$$C_o^{K_1, T} > C_o^{K_2, T}$$

$t=0$ buy $C_{K_1, T}$
 sell $C_{K_2, T}$

Initial Outlay is

$$C_o^{K_1, T} - C_o^{K_2, T}$$

1. If $S_T \leq K_1$ both calls worthless

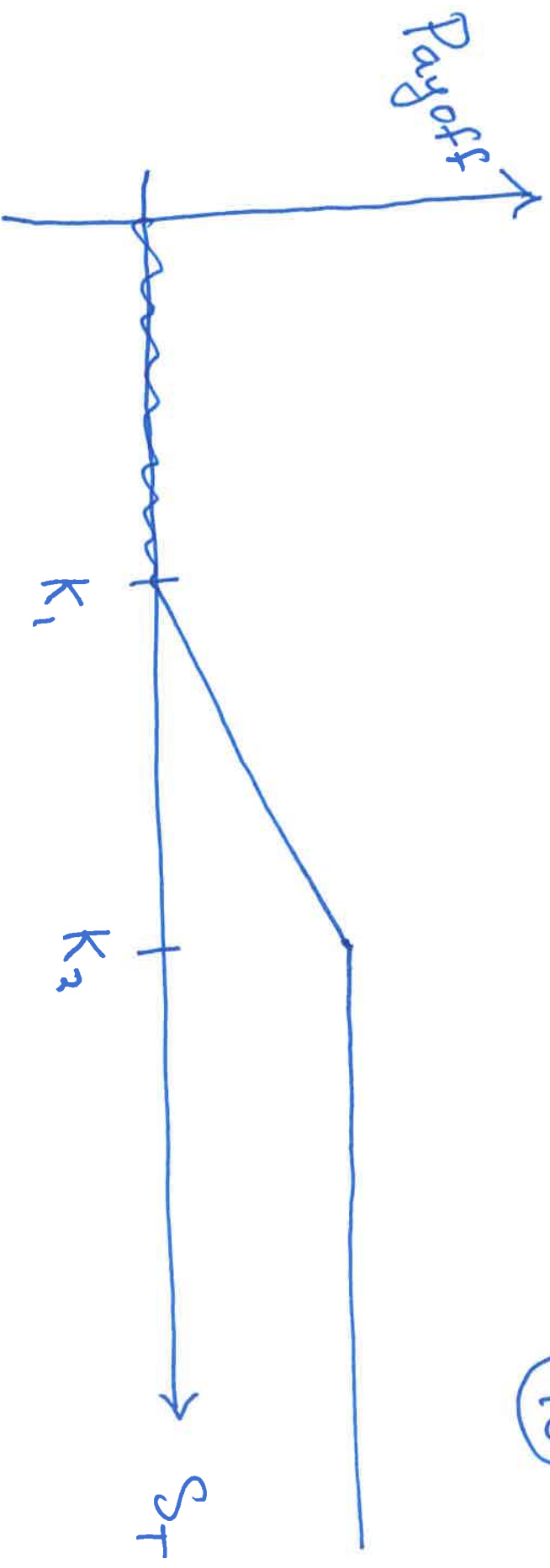
2. If $S_T > K_2$ both calls pay
Net capital at T is $K_2 - K_1$

$$S_T - K_1 - (S_T - K_2) = K_2 - K_1$$

3. If $K_1 \leq S_T \leq K_2$
 K_1 call pays $S_T - K_1$
 K_2 call worthless

Net is $S_T - K_1$

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Exer: Think about Bear Call Spreads
Bull, Bear Put Spreads